॥ श्रीगणेशाय नमः ॥

Numbers

Basic classification of numbers

Base 10 #s: 1234

Face value: 2 (its own)

Place value: 200 (its value in the #)

Place value of zero = 0

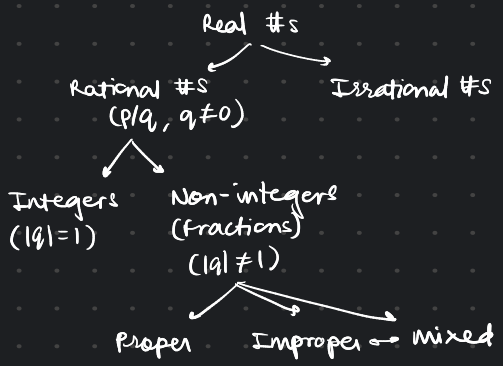
Complex numbers: a + ib

real term, imaginary term

$i = \sqrt{-1}$

a = 0: pure imaginary (ib)

b = 0: real # (a)



Rational numbers:

A number is rational if it can be represented in the form p/q where q ≠ 0, p and q are integers, and p and q have no common factors.

e.g. $\frac{12}{21} \rightarrow \frac{4}{7}$ yes.

Irrational numbers are those that can't be represented in such a form. An irrational number is a non-terminating and non-recurring decimal number.

e.g. $\sqrt{2} = 1.41421356\ldots$, π, etc.

Proper fractions:

$\frac{p}{q}$ $|p| < |q|$ $-1 < \frac{p}{q} < 1$ e.g. $\frac{2}{4} = 0.5$

Improper fractions:

$|p| > |q|$ < -1 or > 1 e.g. $\frac{3}{2} = 1.5$

Mixed fractions:

Improper fractions are also written as mixed fractions.

$\frac{13}{4} = 3\frac{1}{4} = 3.25$

improper, mixed, decimal

Rational #s → Terminating decimal. E.g. $0.54 = \frac{54}{100} = \frac{27}{50}$

Non-terminating decimal

Non-terminating decimals to $\frac{p}{q}$ form:

$0.3333\ldots = 0.\overline{3}$

$x = 0.333\ldots$

$10x = 3.333\ldots$

$9x = 3$

$x = \frac{3}{9} = \frac{1}{3}$

$0.232323\ldots$

$x = 0.\overline{23}$

$100x = 23.\overline{23}$

$x = \frac{23}{99}$

$0.2565656\ldots$

$10x = 2.5656\ldots$

$y = 2.5656\ldots = 10x$

$100y = 256.5656\ldots$

$99y = 254$

$y = 10x = \frac{254}{99}$

$x = \frac{254}{990}$

$0.42323\ldots$

$x = 0.42323\ldots$

$10x = 4.2323\ldots$

$1000x = 423.2323\ldots$

$990x = 423 - 4$

$x = \frac{419}{990}$

Natural #s : +ve integers $\{1, 2, 3, \dots\}$
Whole #s : non-ve integers $\{0, 1, 2, 3, \dots\}$ = Natural #s $\cup \{0\}$.

Properties :

even + even = even
odd + odd = even
even + odd = odd

$(\text{even})^{*} = \text{even}$
$(\text{odd})^{*} = \text{odd}$

odd × odd = odd
∞ × even = even

Prime #s:
A number (other than 1) is prime if it is fully divisible only by 1 or itself. E.g. 17 = 1 × 17 (prime factorisation)
1 is neither prime nor composite.

Coprimes/ relatively primes:
2 #s are coprimes if they only have 1 as a common factor.

E.g. 32 and 27 are both non-primes but they are co-primes.

NOTE:
∞, 1 are coprimes.
prime, prime are coprimes.
(K), (K+1) are coprimes.
sum of 2 coprimes, their product are coprimes.

Twin primes: 2 primes that differ by 2. E.g. (3, 5), (41, 43), (5, 7), (11, 13), etc.

Random shit :
1-15 has 15 primes, 1-100 has 25, 1-200 has 46.
101, 103, 107, 109, 113 are primes.
Largest 3-digit prime : 997
Smallest 4-digit prime : 1009

To check if p is prime :
Let n be smallest counting # s.t. $n^2 \geq p$.
Check if p is divisible by any of the primes $\leq n$.

E.g. 137. $12^2 = 144$.
primes $\leq 12 = \{2, 3, 5, 7, 11\}$ ⇒ 137 is prime.
× × × × ×

If given a prime #, $p^2 - 1$ is always divisible by 24. We can only check non-primes.

Q. $3^{193} + 5$. Prime/not ?

$3^1 = 3$
$3^2 = 9$
27
81
−3

3^{193} = odd
odd + odd = even ← not prime.

Q. 1000001.
→ Prime.
$(10^2)^3 + 1^3$ Divisible by 101.

$a^n + b^n$ is divisible by $a+b$ if n is odd.
$a^n + b^n$ is never divisible by $a-b$.
$a^n - b^n$ is div. by $a-b$ (always). $a^n - b^n$ is div. by $a+b$ when n is even.

Q. $2^{3007}+1$. Is it prime?

→ $a^n + b^n$ is divisible by $(a+b)$.

$2^{3007} + 1^{3007}$ is div. by 3. NON-prime.

Divisibility tests

- By 2: If unit digit $\in \{0, 2, 4, 6, 8\}$.
- By 3: If sum of digits is div. by 3.
- By 4: If # formed by last 2 digits is div by 4.
- By 8: If # formed by last 3 digits is div. by 8.
- By 10: Unit digit = 0
- By 5: Unit digit = 5/0.
- By 6: It is div. by 2 and 3.
- By 9: Sum div. by 9.
- By 7: If diff betn twice the unit digit of given # and the remaining pt. is a multiple of 7 or equal to 0.

#=2576

$2576 \rightarrow 257 - 12 = 245 \leftarrow 24 - 10 = 14$ ← Div. by 7.

- By 11: Diff. betn sum of digits at odd places and that of even places is div by 11 or is zero.
- By 13: 4*Last digit + rest of the # = div. by 13.

$169 \rightarrow 16 + 36 = 52 \rightarrow 5 + 8 = 13$ ✓✓

- By 17: Rest − 5 * unit digit = div. by 17

$612 \rightarrow 61 - 10 = 51 \rightarrow 5 - 5 = 0$ ✓✓

- By 19: 2 * last digit + rest = div. by 19.

Cyclicity : focused on unit digits ; every unit digit has a repetition pattern when raised to a power.

	Unit digits
$2^1 = 2$	2
$2^2 = 4$	4
$2^3 = 8$	6
$2^4 = 16$	8
$2^5 = 32$	2
$2^6 = 64$	4
$2^7 = 128$	6
$2^8 = 256$	8
$2^9 = 512$	2
$2^{10} = 1024$	4

repetitions.

cyclicity = 4.

$3^1 = 3$	7	8
9	49	64
27	343	512
81	2401	4096
243	7	8
729	9	4
7	3	2
1	1	6
cyclicity = 4	c = 4	c = 4

For 2, 3, 7, 8 → cyclicity = 4

For 5, 6, 11 → cyclicity = 1

Q. Unit digit of $12^{12} \times 22^{22}$

→

2
4
8
6
2

12^{12} ↳ rem $12 \% 4 = 0$

⇒ 12^{12} ends with 6

Q. $11^{11} + 12^{12} + 13^{13} + 14^{14} + 15^{15}$ unit digit?

→ $1 + 6 + 3 + 6 + 5 = 21$

Remainder Theorem

remainder of $\frac{a \times b \times c}{n}$ = remainder of $\frac{a_n \times b_n \times c_n}{n}$ where a_n = rem of $\frac{a}{n}$ and so on.

e.g. rem of $\frac{15 \times 17 \times 19}{7}$ = rem of $\frac{1 \times 3 \times 5}{7}$ = rem of $\frac{15}{7}$ = 1

Polynomial Theorem

Accd. to Polynomial theorem,

$$(x+a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}a^1 + {}^nC_2 x^{n-2}a^2 + {}^nC_3 x^{n-3}a^3 + \cdots + {}^nC_{n-1} x^1 a^{n-1} + {}^nC_n a^n$$

rem of $\frac{(x+a)^n}{x}$ = rem of $\frac{a^n}{x}$

Q. Find the remainder of $\frac{9^{99}}{8}$.

→ $\frac{(8+1)^{99}}{8}$ = rem of $\frac{1}{8}$

Q. Rem. of $\frac{5^{100}}{7}$

→ $\frac{5^{100}}{7} = \frac{(5^2)^{50}}{7} = \frac{(25)^{50}}{7}$

$\frac{(3\times7+4)^{50}}{7} = \frac{4^{50}}{7} = \frac{(2^2)^{50}}{7}$

$= \frac{2^{100}}{7} = \frac{(2^3)^{33}\cdot 2}{7} = \frac{8^{33}\cdot 2}{7}$

$= \frac{(7+1)^{33}\cdot 2}{7} = \frac{1^{33}\cdot 2}{7} = \frac{2}{7}$

rem = 2

Squaring #s

$1^2 = 1$	$6^2 = 36$	$11^2 = 121$
$2^2 = 4$	$7^2 = 49$	$12^2 = 144$
$3^2 = 9$	$8^2 = 64$	$13^2 = 169$
$4^2 = 16$	$9^2 = 81$	$14^2 = 196$
$5^2 = 25$	$10^2 = 100$	$15^2 = 225$

Prime Factorization

of factors: $36 = 2^2 \cdot 3^2$

- # of factors $= (2+1)(2+1) = 9$
- # of odd factors $= (2+1) = 3$
- # of even factors $= 9 - 3 = 6$

$360 = 2^3 \cdot 3^2 \cdot 5^1$

of even factors $= (3)(2+1)(1+1) = 18$

↳ NOT considering 2^0

of factors $= (3+1)(2+1)(1+1) = 24$

of odd factors $= (2+1)(1+1) = 6$

Q. How many ways to represent 12 as a product of 2 #s?

$12 = 2^2 \cdot 3^1$ # of factors $= (2+1)(1+1) = 6$

factors $= \{1, 2, 3, 4, 6, 12\}$

3 ways.

Q. # of ways 16 can be reps. as product of 2 #s?

$\rightarrow 16 = 2^4$

of factors $= 5$

of ways $= 3$



of ways $= \lceil \frac{\text{\# of factors}}{2} \rceil$

Sum of factors: $2^2 \cdot 3^1 \cdot 5^6$

$$\left(\frac{2^{2+1}-1}{2-1}\right) \cdot \left(\frac{3^{1+1}-1}{3-1}\right) \cdot \left(\frac{5^{6+1}-1}{6-1}\right)$$

$p_1^{k_1} p_2^{k_2} \dots p_n^{k_n}$

Sum of factors $= \sum_{i=1}^{n} \frac{p_i^{k_i+1}-1}{p_i - 1}$

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of coprimes of n less than n:

$12 \rightarrow (1, 5, 7, 11)$

$n = a^p b^q c^r d^s$

of coprimes $< n$

$= n\left(1-\frac{1}{a}\right) \cdot \left(1-\frac{1}{b}\right)\left(1-\frac{1}{c}\right)\left(1-\frac{1}{d}\right)$

$12 = 2^2 \cdot 3$

multiples of 2: 2, 4, 6, 8, 10

multiples of 3: 3, 6, 9

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\} - \{2, 4, 6, 8, 10\} - \{3, 6, 9\}$

$\{1, 5, 7, 11\}$

$\# = \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right) \cdot 12$

$= \frac{1}{2} \cdot \frac{2}{3} \cdot 12 = 4$

sum of all coprimes of n less than itself $= n \times \frac{(\text{\# of those coprimes less than } n)}{2}$

Factorials

$12! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 \cdot 11 \cdot 12$

How many powers of 2 in 12!?

$12! = 1 \times \underline{2^1} \times 3^1 \times \underline{2^2} \times 5^1 \times \underline{2^1 \times 3^1} \times 7^1 \times \underline{2^3} \times 3^2 \times \underline{2 \times 5} \times 11^1 \times \underline{2^2} \times 3^1$

$12! = 2^{10} \cdot 3^5 \cdot 5^2 \cdot 7 \cdot 11$

Highest power of 2 in 60!?

2	60
2	30
2	15
2	7
2	3
	1

$30 + 15 + 7 + 3 + 1 = 56$

Highest power of 3?

3	60
3	20
3	6
	2

$20 + 6 + 2 = 28$

3^{28}

5	60
5	12
	2

14

$60! = 2^{56} \times 3^{28} \times 5^{14} \times \dots$

Power of 2: $\left\lfloor \frac{60}{2} \right\rfloor + \left\lfloor \frac{30}{2} \right\rfloor + \left\lfloor \frac{15}{2} \right\rfloor + \left\lfloor \frac{7}{2} \right\rfloor + \left\lfloor \frac{3}{2} \right\rfloor + \left\lfloor \frac{1}{2} \right\rfloor$ → 0

$30 + 15 + 7 + 3 + 1 + 0$

$= \underline{56}$

Highest power of 6 in 60!? (non-prime)

Can't get with the same method.

$6 = 2 \times 3$

↳ # of factors would be greater.

$2^{56} \cdot 3^{28} = 2^{28} \underbrace{(2 \times 3)^{28}}_{6^{28}}$

6^{28}

Highest power of 12 that divides 80!

$12 = 2^2 \cdot 3$

2	80
2	40
2	20
2	10
2	5
2	2
	1

→ 78

$2^{78} = (2^2)^{39}$

3	80
3	26
3	8
	2

→ 36

(3^{36})

$(2^2 \times 3)^{36}$ 36

Trailing zeros:

$5! = 120$

↳ # of trailing zeros = 1

From 5! to 9! → 1 trailing zero

of trailing zeros in n! = highest power of 5 in n!

$5! \rightarrow \left\lfloor \frac{5}{5} \right\rfloor + \left\lfloor \frac{1}{5} \right\rfloor$

$1 + 0 = 1$

$10! \rightarrow \left\lfloor \frac{10}{5} \right\rfloor + \left\lfloor \frac{2}{5} \right\rfloor$

$= 2 + 0 = 2$

$10! = 3628800$

Q: Find highest power of 49 in 50! − 49!

$50! - 49! = 50 \times 49! - 49!$

$= 49!(50 - 1)$

$= \underbrace{49}_{1} \times 49!$

1 + power of 49 in 49!

$\therefore$ total # = $1 + 4 = 5$

$49 = 7^2$

7	49
7	7
	1

8

power of 7 in 49! = 8

power of 49 in 49! = $\frac{8}{2} = 4$

↳ 7^2

Q. Find the highest power of 5 in the product of 1st 50 multiples of 5.

→ $(5\times1)\times(5\times2)\times(5\times3)\times(5\times4)\times\cdots\times(5\times50)$

$= 5^{50} \times 50!$

$50 + 12$

$= 62$

5	50
5	10
	2

12

Q. Highest power of 8 in 150! ?

→ $8 = 2^3$

2	150
2	75
2	37
2	18
2	9
2	4
2	2
	1

146

$\frac{146}{3} = 48$

Base conversions

Common systems:
Binary, ternary/trinary, quaternary, quinery, senary, septenary, octal, nonery, decimal, hexadecimal;
(2) (3) (4) (5) (6) (7) (8) (9) (10) (16)

Base B to decimal(10):

$$(\underset{3}{a}\,\underset{2}{b}\,\underset{1}{c}\,\underset{0}{d}\cdot\underset{-1}{e}\,\underset{-2}{f}\,\underset{-3}{g})_B \rightarrow (B^3 a + B^2 b + B^1 c + B^0 d + B^{-1} e + B^{-2} f + B^{-3} g)_{10}$$

E.g. $(123.41)_5 = 25\times1 + 5\times2 + 3 + \frac{1}{5}\times4 + \frac{1}{25}\times1 = 38 + \frac{4}{5} + \frac{1}{25} = 38\frac{21}{25} = (38.84)_{10}$

Decimal to base B:

$(149)_{10}$ → Base 7

7 | 149	
7 | 21	rem = 2
7 | 3	rem = 0
0	rem = 3

$(149)_{10} = (302)_7$

$(18)_{10}$ → Binary

2 | 18	
2 | 9	0
2 | 4	1
2 | 2	0
2 | 1	0
0	1

$(18)_{10} = 10010$ ← divide and put remainders in this direction

Q: $(357)_{10}$ → Base 4

4 | 357	
4 | 89	1
4 | 22	1
4 | 5	2
4 | 1	1
0	1

$(11211)_4$

Q: $(365)_{10}$ → Base 3

3 | 365	
3 | 121	2
3 | 40	1
3 | 13	1
3 | 4	1
3 | 1	1
0	1

$(111112)_3$

Base B_1 to B_2: B_1 → Decimal → B_2

$(341)_5$ → Base 7

$(341)_5 = 25\cdot3 + 5\cdot4 + 1 = (96)_{10} = (165)_7$

7 | 96	
7 | 13	5
7 | 1	6
0	1

Adding:

$$\begin{array}{r} {\scriptstyle 1\,1} \\ (432)_7 \\ +\ (355)_7 \\ \hline (1120)_7 \end{array}$$

8 is 10 in base 7; $(9)_{10} = (12)_7$

7 exceeds base 7 {0,1,2,3,4,5,6}

0 and carry 1 (as 10 is 7)

Q: $(143)_5 + (324)_5$

$$\begin{array}{r} {\scriptstyle 1\,1} \\ 143 \\ +\ 324 \\ \hline (1022)_5 \end{array}$$

Base 5	Base 10
1	1
2	2
3	3
4	4
10	5
11	6
12	7
13	8

Practice questions

Q. Find the # of zeros at the end of the product: 10*20*30*...*200.

→ $(10\times1)\cdot(10\times2)\cdots(10\times20)$

$10^{20}\times(20!)$

$20+4=24$

In 20! → highest power of 5

$10=2\times5$

$=\left\lfloor\frac{20}{5}\right\rfloor+\left\lfloor\frac{20}{5^2}\right\rfloor+\left\lfloor\frac{20}{5^3}\right\rfloor+\cdots$

$=4$

Q. Find the # of zeros at the end of the product: 5*10*15*20*25*30*...*200.

→ $5^{40}(1\times2\times3\times\cdots\times40)$

$5^{40}\times(40!)$

$5^{49}\cdot2^{38}\cdot\dots$

of trailing zeroes = 38.

$\left\lfloor\frac{40}{5}\right\rfloor+\left\lfloor\frac{40}{5^2}\right\rfloor$

$8+1=9$

$\left\lfloor\frac{40}{2}\right\rfloor+\left\lfloor\frac{40}{4}\right\rfloor+\left\lfloor\frac{40}{8}\right\rfloor+\left\lfloor\frac{40}{16}\right\rfloor+\left\lfloor\frac{40}{32}\right\rfloor$

$=20+10+5+2+1$

$=38$

* The # of 2's can be less than # of 5's.

Q. Find the # of zeros at the end of the product: (1^1)*(2^2)*...*(49^49).

→ $1^1\cdot2^2\cdot3^3\cdot4^4\cdots49^{49}$

$5^5\cdot10^{10}\cdot15^{15}\cdot20^{20}\cdot25^{25}\cdot30^{30}\cdot35^{35}\cdot40^{40}\cdot45^{45}$

$5 \quad 5\times2 \quad 5\times3 \quad 5\times4 \quad 5\times5 \quad 5\times6 \quad 5\times7 \quad 5\times8 \quad 5\times9$

Here # of 2's will obv. be more.

$5+10+15+20+25+25+30+35+40+45=250$

Negative remainders:

$17\div9$: $17=9\times1+\underset{\text{rem}}{8}=(9\times2)+\underset{\text{-ve rem}}{(-1)}$

-ve rems range from -1 to $-(n-1)$

Q. $\frac{25\times35\times45\times65\times75}{11}$. Rem = ?

→ $\frac{3\times2\times1\times(-1)\times(-2)}{11}=\frac{12}{11}$ rem = 1

Q. $\frac{-31}{7}$. Rem = ?

→ $7\times(-5)+\underset{\text{rem}=4\text{ or }-3}{4}=-31=7\cdot(-4)+(-3)$

Q. $\frac{(5)^{100}}{7}$. Rem?

→ $\frac{(-2)^{100}}{7}=\frac{(2)^{100}}{7}=\frac{2\times(2^3)^{33}}{7}=\frac{2\times(8)^{33}}{7}=\frac{2\times(1)^{33}}{7}=\frac{2}{7}=2$

Q. $\frac{3^{250}}{7}$. Rem = ?

→ $\frac{3^{250}}{7}\to\frac{9^{125}}{7}\to\frac{2^{125}}{7}\to\frac{2^2\cdot(2^3)^{41}}{7}\to\frac{2^2\cdot8^{41}}{7}\to\frac{2^2\cdot1^{41}}{7}\to\frac{4}{7}\to4$

using -ve rem: $\frac{3^{250}}{7}$ $\quad 3^3=27=\underset{\text{rem}-1}{28-1}$

$\hookrightarrow\frac{3\cdot3^{249}}{7}\to\frac{3\cdot(27)^{83}}{7}\to\frac{3(-1)^{83}}{7}\to\frac{-3}{7}$ rem = −3 = 4

Q. $\frac{(21)^{53}}{17}$ rem = ?

→ $\frac{4^{53}}{17}$ $\quad 4^2 = 16 = 17 - 1$

$\frac{4 \cdot 4^{52}}{17} \rightarrow \frac{4 \cdot (16)^{26}}{17} \rightarrow \frac{4(-1)^{26}}{17} \rightarrow \frac{4}{17}$ $\quad$ rem = 4

Q. Find the remainder when 1!+2!+3!+...+100! is divided by 5.

→ $\frac{1! + 2! + 3! + \cdots + 100!}{5}$

$= \frac{1 + 2 + 6 + 24 + (120 + 720 + \cdots}{5}$ ← all will given rem 0

= rem of $\frac{1+2+6+24}{5} = \frac{3}{=}$

Q. Find the remainder when 1!+2!+3!+...+100! is divided by 6.

→ $1 + 2 + (6 + 24 + 120 + 720 + \cdots$ ← rem = 0 as 3! = 6

$\frac{3}{6}$ → rem = 6

Q. Find the remainder when 1*1!+2*2!+3*3!+...+100*100! is divided by 5.

→ $\frac{1 \times 1! + 2 \times 2! + 3 \times 3! + 4 \times 4! + \cdots + 100 \times 100!}{5}$

$\frac{1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4! + (5 \times 5! + \cdots}{5}$ ← rem = 0

$= \frac{1 + 4 + 18 + 96}{5}$

rem = 4

Q. Find the remainder when ~~1*1!+2*2!+3*3!+...+100*100!~~ is divided by 6.

→ $\frac{1 \cdot 1! + 2 \cdot 2! + (3 \cdot 3! + 4 \cdot 4! + \cdots}{6}$ ← rem = 0

$= \frac{1 + 4}{6}$ rem = 5

Q. Find the remainder when 1*1!+2*2!+3*3!+...+100*100! is divided by 73.

→ $\frac{1 \cdot 1! + 2 \cdot 2! + \cdots + 100 \cdot 100!}{5}$ gives rem 4

by 6 gives rem 5

⋮

by 73 gives rem 72.

$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + 100 \cdot 100!$

$= (2-1)1! + (3-1) \cdot 2! + \cdots + (101-1)\, 100!$

$= (2! + 3! + \cdots + 101!) - (1! + 2! + \cdots + 100!)$

$= 101! - 1!$

NOTE: $\sum_{i=1}^{n} i(i!) = (n+1)! - 1$

Rem of $\frac{(n+1)! - 1}{d} = d-1$ when $d \le n+1$.

↳ $= \frac{1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n!}{d}$

Q. Rem when $\frac{15^{23} + 23^{23}}{19}$.

→ $\frac{(-4)^{23} + (4)^{23}}{19} = 0$

Q. Rem when $\frac{21^{41}}{15}$

→ $\frac{6^{41}}{15} \to \frac{6 \times (6^2)^{20}}{15} \to \frac{6 \times (36)^{20}}{15} \to \frac{6 \times (6)^{20}}{15} \cdots$

Instead,

$\frac{21 \times (21)^{40}}{15} \to$ Div by 3 $\to \frac{7 \times (21)^{40}}{5} \to \frac{2 \cdot (1)^{40}}{5} \to$ rem = 2

But we divided by 3.

So rem = $2 \times 3 = 6$.

Q. Find the last digit of $(2013)^7 + (2015)^9 + (2017)^{11} + (2019)^{13}$

→

$(2013)^7$	$(2015)^9$	$(2017)^{11}$	$(2019)^{13}$
3^2	5^1	7^3	9^1
↓	↓	↓	↓
7	5	3	9

cyclicity of 3 = 4; cyclicity of 5 = 1

$\frac{11}{4}$ rem = 3 (3)

$= 4$

Q. Find the smallest prime # dividing the sum: $3^{11} + 5^{13}$?

→ $3^{11} \to odd^{odd} = odd$

My $5^{13} \to odd$.

odd + odd = even. Div. by 2.

$= 2$

Q. Given: x and y are integers and if x^2/y^3 is an even integer then which of the following must be an even integer?

a) $x-y$ b) $y+1$ c) $\frac{x^2}{y^4}$ d) xy

→ $\frac{x^2}{y^3} = 2k$

$x^2 = y^3 \cdot 2k$

⇒ x^2 is even.

⇒ x is even.

⇒ xy is even.

∴ d.

can't say anything about the rest.

Q. How many zeroes are there if we multiply all the prime #s between 0 and 200.

→ $2 * 3 * 5 * 7 * 11 * \ldots$

one zero

rest obv. won't be multiples of 2/5 as they are prime.

Q. N = n(n+1)(n+2)(n+3)(n+4), where n is a natural #. Which of the following statements is/are true?
a. Unit digit of N is 0. b. N is perfectly divisible by 24. c. N is a perfect square. d. N is odd.

→ $n(n+1)(n+2)(n+3)(n+4)$

5 consecutive. ∴ we'll always get 1 multiple of 5 and 1 multiple of 2.

⇒ opt. a is true.

Also we'll get 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5

A multiple of 8 and a multiple of 3.

⇒ div. by 24.

⇒ opt. b is true.

$1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$ NOT a perfect square.

Q. 2419! = (504)^a * b. b is not a multiple of 7. Find a.

→ $\left\lfloor \frac{2419}{7} \right\rfloor + \left\lfloor \frac{2419}{49} \right\rfloor + \left\lfloor \frac{2419}{343} \right\rfloor + \left\lfloor \frac{2419}{2401} \right\rfloor$

$= 345 + 49 + 7 + 1$

$= 402$

$504 = 2^3 \cdot 3^2 \cdot 7$

b doesn't have 7.

⇒ $(504)^a$ fulfils all > powers ie. $(504)^a$ has 7^{402} (has 7^9)

⇒ a = 402

Q. How many factors of N are multiples of K?

$N = 12^{12} \cdot 14^{14} \cdot 15^{15}$

$K = 12^{10} \cdot 14^{10} \cdot 15^{10}$

→ $N = 2^{38} \times 3^{27} \times 5^{15} \times 7^{14}$

Total # of factors $= 39 \times 28 \times 16 \times 15$

$K = (3 \cdot 2^2)^{10} \cdot (2 \cdot 7)^{10} \cdot (3 \cdot 5)^{10} = 2^{30} \cdot 3^{20} \cdot 5^{10} \cdot 7^{10}$

Total # of factors $= 31 \cdot 21 \cdot 11 \cdot 11$

$\frac{N}{K} = \frac{2^{38} \cdot 3^{27} \cdot 5^{15} \cdot 7^{14}}{2^{30} \cdot 3^{20} \cdot 5^{10} \cdot 7^{10}} = 2^8 \cdot 3^7 \cdot 5^5 \cdot 7^4$

of factors $= 9 \cdot 8 \cdot 6 \cdot 5$

Q. A man wrote all the natural #s starting from 1. What will be the 50th digit of the number?

→ 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

11 12 13 14 15 16 17 18 19 20 21 22 23

50th digit: 9 are single digits.

41 left

10–29 gives 40 next digits.

8 9 10 11 12 ⋯ 29 30 31

9 10 11 +9

50th digit

Q. Which of these has a total of 24 factors?

a) 21×2^3 b) $2^7 \times 12^3$ c) $2^6 \cdot 3^4$ d) 63×55

→

a) $= 2^3 \cdot 3^1 \cdot 7^1$

$4 \cdot 2 \cdot 2 = 16$

b) $= 2^7 \cdot 2^6 \cdot 3^3$

$= 2^{13} \cdot 3^3$

$14 \cdot 4 = 56$

c) $7 \cdot 5 = 35$

d) $= 21 \times 3 \times 11 \times 5$

$= 3^2 \cdot 5 \cdot 7 \cdot 11$

$3 \cdot 2 \cdot 2 \cdot 2 = 24$

Q. What is the highest power of 5 that divides 1001*1002*...*2010?

→

1005 | 5, 10, 15, 20, ..., 1000

1010 | 1, 2, 3, 4, ..., 200

1015 |

1020 |

⋮ |

1995 | → 200

2000 |

2005 |

2010 | → 2

202 + those that have more than one 5.

1025 | 25 50 75 100 ... 1000

1050 | 1 2 3 ... 40

1075 |

⋮ |

multiples of 25 2000 | → 40

1125 |

1250 |

1375 |

1500 | → 8

1625 |

1750 |

1875 |

2000 |

of 125

1250 |

1875 | → 2

of 625

202 + 40 + 8 + 2 = 252

OR

given product is $\frac{2010!}{1000!}$

powers of 5 in 2010! − powers of 5 in 1000! = 252

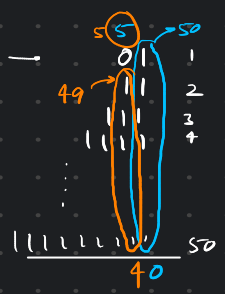
Q. If 45376p is divisible by 3 then minimum value of p is ___?

→ $4+5+3+7+6+p$

$= 25+p$

$p = 2$

Q. What is the tens digit of the sum of the first 50 terms of 1, 11, 111, 1111, ... ?



→ 4

Q. what is the remainder when $N = \frac{(1!+2!+\dots+1000!)^{40}}{10}$?

→ $1!+2!+3!+4!+\underbrace{5!+6!+7!+\dots}$

all will have 10 as multiple.

$\frac{1+2+6+24}{10}$ $\quad \frac{(33)^{40}}{10} \rightarrow \frac{3^{40}}{10}$ Div. by 10 is just unit digit of 33^{40}.

$40 \div 4 \rightarrow$ rem $= 0$

Unit digit $= 1$

3	1
9	2
7	3
1	0
3	

Ans = 1

Q. Rem of $\frac{9^{99}}{7}$. Alt method: using cyclicity

→ rem. of $\frac{9^1}{7} = 2$

$\frac{9^2}{7} \rightarrow \frac{9 \cdot 9}{7} \rightarrow \frac{2 \cdot 2}{7} \rightarrow 4$

$\frac{9^3}{7} \rightarrow \frac{2 \cdot 2 \cdot 2}{7} \rightarrow 1$

$\frac{9^4}{7} \rightarrow \frac{1 \cdot 9}{7} \rightarrow 2$

$\frac{9^5}{7} \rightarrow \frac{2 \cdot 9}{7} \rightarrow 4$

$\frac{9^6}{7} \rightarrow 1$

∴ cyclicity = 3

2	$9^1/7$
4	$9^2/7$
1	$9^3/7$
2	
4	
1	
⋮	

$\frac{99}{3} \rightarrow$ rem $= 0$

$\frac{9^{99}}{7} \rightarrow$ rem $= 1$

Q. Rem of $\frac{9^{71}}{7}$ → $\frac{71}{3} \rightarrow$ rem $= 2$ ⇒ $\frac{9^{71}}{7} \rightarrow$ rem $= 4$

Q. Rem of $\frac{7^{77}}{15}$

→ $\frac{7^1}{15} \rightarrow 7$

$\frac{7^2}{15} \rightarrow 4$

$\frac{7^3}{15} \rightarrow \frac{7 \cdot 4}{15} \rightarrow -2 \rightarrow 13$

$\frac{7^4}{15} \rightarrow \frac{(-2)(7)}{15} \rightarrow -14 \rightarrow 1$

$\frac{7^5}{15} \rightarrow \frac{1 \cdot 7}{15} \rightarrow 7$

$\frac{7^6}{15} \rightarrow 4$

cyclicity = 4.

$\frac{77}{4} \rightarrow$ rem $= 1$

rem of $\frac{7^{77}}{15} \rightarrow 7$

Q. If 554 in base b is equal to square of the # 24 in base b, what's b?

→ $(24)_b^2 = (554)_b$

$\Rightarrow (2b+4)^2 = 5b^2+5b+4$

$4b^2+16b+16 = 5b^2+5b+4$

$b^2-11b-12=0$

$(b-12)(b+1)=0$

$b=12$

Q. p is a prime #. The given # is completely divisible for ___ values of p? $\frac{10000!}{p^p}$

→ for 101: max power $= 101^{99}$

101	10000
101	99
	0

101^{101} does not div.

for 97:

97	10000
97	103
	1

97^{109}

$\therefore 97^{97}$ divides 10000!

So 97 is the max value.

of primes $2, 3, 5, 7, \ldots, 97 \rightarrow 25$.

Q. n is a natural #. How many values of n are possible such that $\frac{6n^3+3n^2+n+24}{n}$ is also a natural #.

→ $6n^2+3n+1+\frac{24}{n}$

$\frac{24}{n}$ has to be a natural #. $n = \{1, 2, 3, 4, 6, 8, 12, 24\}$. or $24 = 2^3 \cdot 3$

$4 \cdot 2 = 8$

$= 8$

Q. Let x and y be +ve integers s.t. x is prime and y is composite. Then
a. y-x cannot be even.
b. (x+y)/x cannot be even.
c. (x+y) cannot be even.
d. None of these. ✓✓

→ a) $4-2=2$ False

b) $\frac{2+10}{2}=6$ False

c) $2+4=6$ False

Q. In a certain base, 137+254=402. What is 342+562 in the same base?

→

$$\begin{array}{r} 137 \\ +\ 254 \\ \hline 402 \end{array}$$

$7+4=11 \rightarrow$ 11 is __2 in that base.

$11 \rightarrow 12$ in base 9.

$$\begin{array}{r} {\scriptstyle 1\ \ \ \ } \\ 342 \\ +\ 562 \\ \hline 1014 \end{array}$$

Q. Find the # of +ve integers which divide 2010 and leaves a remainder of 10. 20

→ 2010 = _ * _ + 10

_ * _ = 2000

$2000 = 2^4 \cdot 5^3$

of factors = $5 \cdot 4 = 20$